

Who Survives a Fatal Crash?

COSC 522 - Machine Learning | Project Proposal

Problem and Motivation

The Fatality Analysis Reporting System (FARS), maintained by the National Highway Traffic Safety Administration, records every U.S. motor-vehicle crash on a public road in which at least one person dies within 30 days. Because FARS contains only fatal crashes, it cannot be used to estimate whether a crash will occur or whether an arbitrary crash will become fatal. Instead, our project asks a conditional question:

Given that a crash was severe enough to kill at least one person, what predicts whether a particular vehicle occupant survives?

Not everyone involved in a fatal crash dies. FARS records injury outcomes for all people involved, allowing us to study differences between occupants who survive and those who do not within this extreme subset of crashes. The problem is especially relevant to occupant protection because factors such as restraint use, airbag deployment, seating position, occupant age, and vehicle characteristics may help explain why people involved in similarly severe crashes experience different outcomes.

This project falls under the course's Novel Application track because it develops a complete machine-learning pipeline for a real-world safety problem using a large public dataset.

Data and Prediction Task

We will use FARS data from 2016 through 2023. FARS contains linked crash-, vehicle-, and person-level information; our unit of analysis will be the individual motor-vehicle occupant.

Non-motorists such as pedestrians and cyclists will be excluded because their outcome distribution within fatal crashes is substantially different from that of vehicle occupants. Records with unknown injury severity or people coded as having died prior to the crash will also be excluded.

The prediction target will be binary: survived versus died from injuries sustained in the crash. Features will combine person-, vehicle-, crash-, and roadway-level information. Pre-crash and vehicle-design features include restraint use, airbag information, seating position, age, sex, vehicle body type, model year, manner of collision, initial impact point, light and weather conditions, land use, roadway functional class, and speeding involvement.

Modeling Approach

A key part of the project is distinguishing variables that describe conditions or protection before and during the collision from variables that are consequences of the crash itself. We will therefore construct two feature sets:

Feature set	Purpose
Pre-crash/design	Uses occupant, vehicle, roadway, environmental, and collision characteristics such as restraint use, seating position, vehicle type, and impact configuration.
Expanded	Adds crash-consequence variables such as ejection, fire occurrence, and extent of vehicle damage.

Downstream variables may substantially improve predictive accuracy, but because they are partly consequences of crash severity, they have a different interpretation from protective factors. Comparing the two models will show how much predictive performance comes from information available independently of those crash consequences.

Our baseline will be logistic regression, which provides an interpretable reference model and is consistent with conventional statistical approaches used in crash-safety research. We will compare it with nonlinear tree-based methods, primarily gradient-boosted trees using XGBoost or LightGBM, with random forests as an additional comparison if appropriate. For the tree-based models, we will use SHAP values to examine which features contribute most strongly to survival predictions.

Generalization Experiments

Beyond standard predictive performance, we will test whether the learned relationships generalize across both geography and time.

For geographic generalization, models will be trained on a subset of states and evaluated on states that were not represented during training. Because FARS uses a common federal coding standard across states, this provides a relatively clean test of whether relationships learned in one set of jurisdictions transfer to others.

For temporal generalization, we will train models using 2016-2020 data and evaluate them on 2021-2023 data. This will test whether the model remains effective on later crashes rather than only on randomly held-out observations from the same period.

Expected Contribution

Most prior work either studies individual crash-risk factors using conventional statistical models or predicts injury severity using datasets containing crashes across a broader range of severity. Our project instead focuses specifically on survival conditional on already being involved in a fatal crash.

The project contributes three elements: a predictive formulation of occupant survival within fatal crashes, an explicit comparison between models using pre-crash/design information and those incorporating crash consequences, and an evaluation of whether the resulting models generalize across unseen states and future years.